Derivation of Liquid Thermal Boundary Layer Under Transient Square-Wave AI Workloads

Step 1: Exact 1D Unperturbed Temperature Field Solution

Transient Flow Boiling Project

August 6, 2026

1 Introduction & Physical Setup

In direct-to-chip two-phase cooling applications, high-performance processors (such as AI accelerators) do not generate uniform, steady-state heat flux. Instead, their power draw switches rapidly between high-power active computation states and low-power idle states. We model this dynamic workload as a periodic square-wave heat flux $q''(t)$ applied at the heater surface.

Before boiling begins (i.e., prior to bubble nucleation), heat transfers from the solid wall into the adjacent liquid purely via transient thermal conduction. We treat the liquid as a semi-infinite medium occupying the spatial region $y \geq 0$, where $y = 0$ corresponds to the heater wall, and $y > 0$ extends perpendicularly into the fluid channel.

To simplify the governing equations, we define the **liquid superheat** (temperature excess relative to the fluid saturation temperature T_{sat}) as:

$$\Theta(y,t) \equiv T(y,t) - T_{\text{sat}} \tag{1}$$

The liquid is initially at rest and uniformly at saturation, meaning $\Theta(y,0) = 0$.

—

2 Governing Equations & Boundary Conditions

Heat transport across the thin liquid boundary layer is governed by the 1D transient heat diffusion equation:

$$\frac{\partial \Theta(y,t)}{\partial t} = \alpha_l \frac{\partial^2 \Theta(y,t)}{\partial y^2}, \quad y > 0,\ t > 0 \tag{2}$$

where $\alpha_l = \frac{k_l}{\rho_l c_{p,l}}$ is the thermal diffusivity of the liquid, k_l is thermal conductivity, ρ_l is density, and $c_{p,l}$ is specific heat capacity.

To solve this partial differential equation (PDE), we require one initial condition and two spatial boundary conditions:

2.1 1. Initial Condition (IC)

Before time $t = 0$, no power is applied to the heater. The liquid domain is completely isothermal at the saturation temperature:

$$\Theta(y,0) = 0 \quad \text{for all } y \geq 0 \tag{3}$$

2.2 2. Wall Boundary Condition (BC 1 at $y = 0$)

At the surface of the heater ($y = 0$), heat enters the liquid domain via Fourier's Law of conduction. The conductive heat flux leaving the surface into the fluid must equal the time-varying heat flux generated by the AI chip $q''(t)$:

$$-k_l \left.\frac{\partial \Theta(y,t)}{\partial y}\right|_{y=0} = q''(t) \implies \left.\frac{\partial \Theta(y,t)}{\partial y}\right|_{y=0} = -\frac{q''(t)}{k_l} \tag{4}$$

2.3 3. Far-Field Boundary Condition (BC 2 as $y \to \infty$)

Deep inside the fluid channel, far away from the heated wall, the liquid remains unperturbed at the bulk saturation temperature:

$$\lim_{y\to\infty} \Theta(y,t) = 0 \quad \text{for all } t \geq 0 \tag{5}$$

—

3 Step 1A: Solution for a Constant Heat Flux Step Input (q''_{peak})

Before tackling a repeating square wave, we solve the fundamental problem of a single, constant heat flux q''_{peak} turned ON at $t = 0$ and maintained indefinitely. We use the Laplace transform method, which converts the time-dependent PDE into an ordinary differential equation (ODE) in space.

3.1 Transforming the PDE to the Laplace Domain

We define the Laplace transform of the temperature excess as:

$$\mathcal{L}\{\Theta(y,t)\} = \bar{\Theta}(y,s) = \int_0^\infty \Theta(y,t)e^{-st}\,dt \tag{6}$$

Applying the time-derivative property of Laplace transforms $\mathcal{L}\left\{\frac{\partial \Theta}{\partial t}\right\} = s\bar{\Theta}(y,s) - \Theta(y,0)$ and using our initial condition $\Theta(y,0) = 0$:

$$s\bar{\Theta}(y,s) = \alpha_l \frac{d^2\bar{\Theta}(y,s)}{dy^2} \tag{7}$$

Rearranging terms yields a second-order linear homogeneous ODE in space:

$$\frac{d^2\bar{\Theta}(y,s)}{dy^2} - \frac{s}{\alpha_l}\bar{\Theta}(y,s) = 0 \tag{8}$$

The general solution to this ordinary differential equation is a combination of decaying and growing spatial exponentials:

$$\bar{\Theta}(y,s) = C_1 e^{-y\sqrt{s/\alpha_l}} + C_2 e^{+y\sqrt{s/\alpha_l}} \tag{9}$$

where C_1 and C_2 are integration constants in space that depend on s.

3.2 Determining Integration Constants

First, we apply the far-field boundary condition (BC 2). In Laplace space, as $y \to \infty$, $\bar{\Theta}(y, s)$ must remain bounded. Because the term $e^{+y\sqrt{s/\alpha_l}}$ grows infinitely as $y \to \infty$, we must set:

$$C_2 = 0 \tag{10}$$

Leaving us with:

$$\bar{\Theta}(y, s) = C_1 e^{-y\sqrt{s/\alpha_l}} \tag{11}$$

Next, we apply the wall boundary condition (BC 1). For a constant step heat flux $q''(t) = q''_{\text{peak}}$, its Laplace transform is $\mathcal{L}\{q''_{\text{peak}}\} = \frac{q''_{\text{peak}}}{s}$. Taking the derivative of our spatial solution with respect to y at $y = 0$:

$$\left.\frac{d\bar{\Theta}}{dy}\right|_{y=0} = -C_1\sqrt{\frac{s}{\alpha_l}} = -\frac{q''_{\text{peak}}}{sk_l} \tag{12}$$

Solving this algebraic equation for C_1:

$$C_1 = \frac{q''_{\text{peak}}\sqrt{\alpha_l}}{k_l s^{3/2}} \tag{13}$$

Substituting C_1 back into our equation gives the exact temperature field in the Laplace domain:

$$\bar{\Theta}(y, s) = \frac{q''_{\text{peak}}\sqrt{\alpha_l}}{k_l}\left(\frac{e^{-y\sqrt{s/\alpha_l}}}{s^{3/2}}\right) \tag{14}$$

3.3 Inverse Laplace Transform back to Time Domain

To return to the time domain t, we use the standard inverse Laplace transform identity:

$$\mathcal{L}^{-1}\left\{\frac{e^{-a\sqrt{s}}}{s^{3/2}}\right\} = 2\sqrt{\frac{t}{\pi}}\exp\left(-\frac{a^2}{4t}\right) - a\,\text{erfc}\left(\frac{a}{2\sqrt{t}}\right) \tag{15}$$

Setting $a = \frac{y}{\sqrt{\alpha_l}}$, the solution becomes:

$$\Theta_{\text{step}}(y, t) = \frac{2q''_{\text{peak}}\sqrt{\alpha_l t}}{k_l}\left[\frac{1}{\sqrt{\pi}}\exp\left(-\frac{y^2}{4\alpha_l t}\right) - \frac{y}{2\sqrt{\alpha_l t}}\text{erfc}\left(\frac{y}{2\sqrt{\alpha_l t}}\right)\right] \tag{16}$$

To make this formula compact, we use the standard mathematical notation for the integral of the complementary error function, defined as $\text{ierfc}(z) \equiv \frac{e^{-z^2}}{\sqrt{\pi}} - z\,\text{erfc}(z)$. Substituting this yields the classic constant step flux solution:

$$\Theta_{\text{step}}(y, t) = \frac{2q''_{\text{peak}}\sqrt{\alpha_l t}}{k_l}\,\text{ierfc}\left(\frac{y}{2\sqrt{\alpha_l t}}\right) \tag{17}$$

—

4 Step 1B: Derivation of a Single Finite Heat Pulse

A single pulse has a duration of t_{on}, meaning heat turns ON at $t = 0$ and turns OFF at $t = t_{\text{on}}$.

Because heat diffusion is a linear partial differential equation, we cannot simply "turn off" the original solution. Instead, we use the **principle of linear superposition**. We model a finite pulse as two independent, continuous heating sources running simultaneously:

1. **Heater A (Positive Step):** A heat flux of $+q''_{\text{peak}}$ that turns ON at $t = 0$ and stays ON forever.

2. **Heater B (Negative Step):** A fictitious "anti-heater" with a heat flux of $-q''_{\text{peak}}$ that turns ON later at $t = t_{\text{on}}$ and stays ON forever.

Mathematically, the net boundary condition is represented using the Heaviside step function $H(\tau)$:

$$q''(t) = \underbrace{q''_{\text{peak}} \cdot H(t)}_{\text{Heater A}} + \underbrace{\left(-q''_{\text{peak}}\right) \cdot H(t - t_{\text{on}})}_{\text{Heater B}} \tag{18}$$

When $t < t_{\text{on}}$, only Heater A is active ($q'' = q''_{\text{peak}}$). When $t \geq t_{\text{on}}$, both heaters are active, so the net flux at the wall cancels out perfectly: $q'' = (+q''_{\text{peak}}) + (-q''_{\text{peak}}) = 0$.

4.1 Applying the Time-Shift Theorem to Heater B

The thermal response to Heater A is simply the step solution derived in Step 1A:

$$\Theta_A(y,t) = \frac{2q''_{\text{peak}}\sqrt{\alpha_l t}}{k_l} \operatorname{ierfc}\left(\frac{y}{2\sqrt{\alpha_l t}}\right) H(t) \tag{19}$$

For Heater B, its activation is delayed by t_{on}. By the Second Shift Theorem of Laplace transforms ($\mathcal{L}^{-1}\{F(s)e^{-as}\} = f(t-a)H(t-a)$), shifting an event in time by t_{on} simply replaces every instance of time t in the solution with the elapsed time since activation, $(t - t_{\text{on}})$:

$$\Theta_B(y,t) = -\frac{2q''_{\text{peak}}\sqrt{\alpha_l (t - t_{\text{on}})}}{k_l} \operatorname{ierfc}\left(\frac{y}{2\sqrt{\alpha_l (t - t_{\text{on}})}}\right) H(t - t_{\text{on}}) \tag{20}$$

4.2 Combined Single-Pulse Solution

Summing Θ_A and Θ_B gives the complete temperature distribution for a single pulse:

$$\Theta_{\text{pulse}}(y,t) = \frac{2q''_{\text{peak}}\sqrt{\alpha_l}}{k_l} \left[\sqrt{t} \operatorname{ierfc}\left(\frac{y}{2\sqrt{\alpha_l t}}\right) H(t) - \sqrt{t - t_{\text{on}}} \operatorname{ierfc}\left(\frac{y}{2\sqrt{\alpha_l (t - t_{\text{on}})}}\right) H(t - t_{\text{on}})\right] \tag{21}$$

—

5 Step 1C: Detailed Explanation of the Periodic Pulse Train Solution

Now we extend the single-pulse concept to a continuous, repeating square wave representing ongoing AI processing workloads.

5.1 Understanding Thermal Memory in Linear Systems

In a physical system governed by linear differential equations, thermal effects propagate outward and linger indefinitely. When a heat pulse turns off at the wall, the thermal energy already injected into the fluid does not vanish; it continues to diffuse deeper into the fluid domain ($y > 0$).

Therefore, when Cycle 2 begins, the liquid does not start from a cold, isothermal state. It carries the residual heat remaining from Cycle 1. This accumulated temperature rise across all past operation cycles is called **thermal memory**. Because the heat diffusion equation is linear, the temperature profile at any present moment t is simply the sum of the thermal responses caused by every historical switching event that has ever occurred prior to t.

5.2 Indexing Switching Events Across Multiple Cycles

Let P be the period of one complete pulse cycle ($P = 1/f$, where f is frequency), and let t_{on} be the ON duration during each cycle.

We index each complete cycle using an integer $n \in \{0, 1, 2, 3, \dots\}$, where $n = 0$ is the very first cycle starting at $t = 0$. For any given cycle n, two discrete switching events take place:

1. **The n-th Turn-ON Event:** Power jumps to $+q''_{\text{peak}}$ at timestamp $t_{n,\text{on}} = nP$.
2. **The n-th Turn-OFF Event:** Power drops to 0 at timestamp $t_{n,\text{off}} = nP + t_{\text{on}}$.

Thus, we can construct the continuous periodic square wave as an infinite series of positive and negative step inputs:

$$q''(t) = q''_{\text{peak}} \sum_{n=0}^{\infty} \Big[\underbrace{H(t - nP)}_{n\text{-th Turn-ON}} - \underbrace{H\big(t - (nP + t_{\text{on}})\big)}_{n\text{-th Turn-OFF}} \Big] \tag{22}$$

6 Step 1C: Detailed Explanation of the Periodic Pulse Train Solution

Now we extend the single-pulse concept to a continuous, repeating square wave representing ongoing AI processing workloads.

6.1 Understanding Thermal Memory in Linear Systems

In a physical system governed by linear differential equations, thermal effects propagate outward and linger indefinitely. When a heat pulse turns off at the wall, the thermal energy already injected into the fluid does not vanish; it continues to diffuse deeper into the fluid domain ($y > 0$).

Therefore, when Cycle 2 begins, the liquid does not start from a cold, isothermal state. It carries the residual heat remaining from Cycle 1. This accumulated temperature rise across all past operation cycles is called **thermal memory**. Because the heat diffusion equation is linear, the temperature profile at any present moment t is simply the sum of the thermal responses caused by every historical switching event that has ever occurred prior to t.

6.2 Indexing Switching Events Across Multiple Cycles

Let P be the period of one complete pulse cycle ($P = 1/f$, where f is frequency), and let t_{on} be the ON duration during each cycle.

We index each complete cycle using an integer $n \in \{0, 1, 2, 3, \dots\}$, where $n = 0$ is the very first cycle starting at $t = 0$. For any given cycle n, two discrete switching events take place:

1. **The n-th Turn-ON Event:** Power jumps to $+q''_{\text{peak}}$ at timestamp $t_{n,\text{on}} = nP$.
2. **The n-th Turn-OFF Event:** Power drops to 0 at timestamp $t_{n,\text{off}} = nP + t_{\text{on}}$.

Thus, we can construct the continuous periodic square wave as an infinite series of positive and negative step inputs:

$$q''(t) = q''_{\text{peak}} \sum_{n=0}^{\infty} \Big[\underbrace{H(t - nP)}_{n\text{-th Turn-ON}} - \underbrace{H\big(t - (nP + t_{\text{on}})\big)}_{n\text{-th Turn-OFF}} \Big] \tag{23}$$

6.3 Mathematical Proof: From Flux Summation to Temperature Summation

To rigorously justify how the sum of heat flux boundary conditions transforms directly into a sum of temperature fields, we invoke the linear operator properties of heat diffusion.

6.3.1 1. PDE Linearity and Time Invariance

Consider the 1D heat conduction operator $\mathcal{L} \equiv \frac{\partial}{\partial t} - \alpha_l \frac{\partial^2}{\partial y^2}$. The governing system is linear:

$$\mathcal{L}[\Theta(y,t)] = 0, \quad -k_l \left.\frac{\partial \Theta}{\partial y}\right|_{y=0} = q''(t) \tag{24}$$

Because $\mathcal{L}$ and the boundary condition operator are linear, the system obeys **linear superposition**: if a flux profile $q''_m(t)$ produces a temperature field $\Theta_m(y,t)$, then a composite flux $q''_{\text{net}}(t) = \sum_m q''_m(t)$ produces the exact composite temperature field $\Theta_{\text{net}}(y,t) = \sum_m \Theta_m(y,t)$.

Furthermore, because thermophysical properties (α_l, k_l) are constant, the governing differential equation is **time-invariant**. A heat flux step $q''_{\text{step}}(t-\tau)$ initiated at a delayed time $t = \tau$ produces the exact same spatial temperature field as one initiated at $t = 0$, shifted in time by τ:

$$\Theta_{\text{step}}(y,t;\tau) = \Theta_{\text{step}}(y,t-\tau)H(t-\tau) \tag{25}$$

6.3.2 2. Mapping Individual Flux Steps to Thermal Responses

Recall from Step 1A that a single step change in heat flux of magnitude $+q''_{\text{peak}}$ applied at $t = 0$ yields the thermal kernel:

$$\Theta_{\text{kernel}}(y,\Delta t) = \frac{2q''_{\text{peak}}\sqrt{\alpha_l}}{k_l}\sqrt{\Delta t}\,\text{ierfc}\left(\frac{y}{2\sqrt{\alpha_l \Delta t}}\right) \tag{26}$$

Applying time invariance, we map each individual component of our square wave flux expansion directly to its exact temperature response:

- **n-th Turn-ON Step ($+q''_{\text{peak}}$ at $t_{n,\text{on}} = nP$):**

$$\Theta_{n,\text{on}}(y,t) = \Theta_{\text{kernel}}(y, t-nP)H(t-nP) \tag{27}$$

- **n-th Turn-OFF Step ($-q''_{\text{peak}}$ at $t_{n,\text{off}} = nP + t_{\text{on}}$):**

$$\Theta_{n,\text{off}}(y,t) = -\Theta_{\text{kernel}}(y, t-nP-t_{\text{on}})H(t-nP-t_{\text{on}}) \tag{28}$$

6.3.3 3. Constructing the Complete Temperature Field Summation

By applying linear superposition across all cycles $n = 0, 1, 2, \ldots$:

$$\Theta(y,t) = \sum_{n=0}^{\infty} \left[\Theta_{n,\text{on}}(y,t) + \Theta_{n,\text{off}}(y,t)\right] \tag{29}$$

We define the relative time elapsed since each switching event:

$$\Delta t_{n,\text{on}} \equiv t - nP, \quad \Delta t_{n,\text{off}} \equiv t - nP - t_{\text{on}} \tag{30}$$

Because $H(\Delta t_{n,\text{on}}) = 0$ for all $n > \lfloor t/P \rfloor$, future events contribute zero to the sum. Truncating the infinite series at the current active cycle index $N(t) = \lfloor t/P \rfloor$ and factoring out common terms yields the final analytical solution:

$$\Theta(y,t) = \frac{2q''_{\text{peak}}\sqrt{\alpha_l}}{k_l}\sum_{n=0}^{N(t)}\left[\sqrt{\Delta t_{n,\text{on}}}\,\text{ierfc}\left(\frac{y}{2\sqrt{\alpha_l \Delta t_{n,\text{on}}}}\right) - \sqrt{\Delta t_{n,\text{off}}}\,\text{ierfc}\left(\frac{y}{2\sqrt{\alpha_l \Delta t_{n,\text{off}}}}\right)H(\Delta t_{n,\text{off}})\right] \tag{31}$$

This completes the proof.

6.4 Constructing the Analytical Temperature Field Summation

By applying linear superposition to this infinite series of switching events, we calculate the net liquid temperature superheat $\Theta(y,t)$ at depth y and current time t.

We introduce the floor function $N(t) = \lfloor t/P \rfloor$, which represents the index of the cycle currently taking place at time t. Any switching event scheduled to occur at a time greater than t (in the future) has an elapsed time that is negative, causing its Heaviside step function $H(\tau)$ to equal zero. This naturally truncates the infinite mathematical series into a finite sum over all active and past cycles $n = 0, 1, \ldots, N(t)$:

$$\Theta(y,t) = \frac{2q''_{\text{peak}}\sqrt{\alpha_l}}{k_l} \sum_{n=0}^{N(t)} \left[\sqrt{\Delta t_{n,\text{on}}}\, \text{ierfc}\left(\frac{y}{2\sqrt{\alpha_l \Delta t_{n,\text{on}}}} \right) - \sqrt{\Delta t_{n,\text{off}}}\, \text{ierfc}\left(\frac{y}{2\sqrt{\alpha_l \Delta t_{n,\text{off}}}} \right) H(\Delta t_{n,\text{off}}) \right] \tag{32}$$

where:

- $\Delta t_{n,\text{on}} = t - nP$ is the total elapsed time since the n-th pulse turned ON.
- $\Delta t_{n,\text{off}} = t - nP - t_{\text{on}}$ is the total elapsed time since the n-th pulse turned OFF.
- $H(\Delta t_{n,\text{off}})$ is a step function that equals 1 if the n-th pulse has already turned OFF at current time t, and 0 if the n-th pulse is currently still in its ON phase.

This closed-form summation provides an exact mathematical model of the 1D liquid thermal boundary layer $T(y,t) = T_{\text{sat}} + \Theta(y,t)$ prior to bubble nucleation under any arbitrary, periodic AI heating profile.